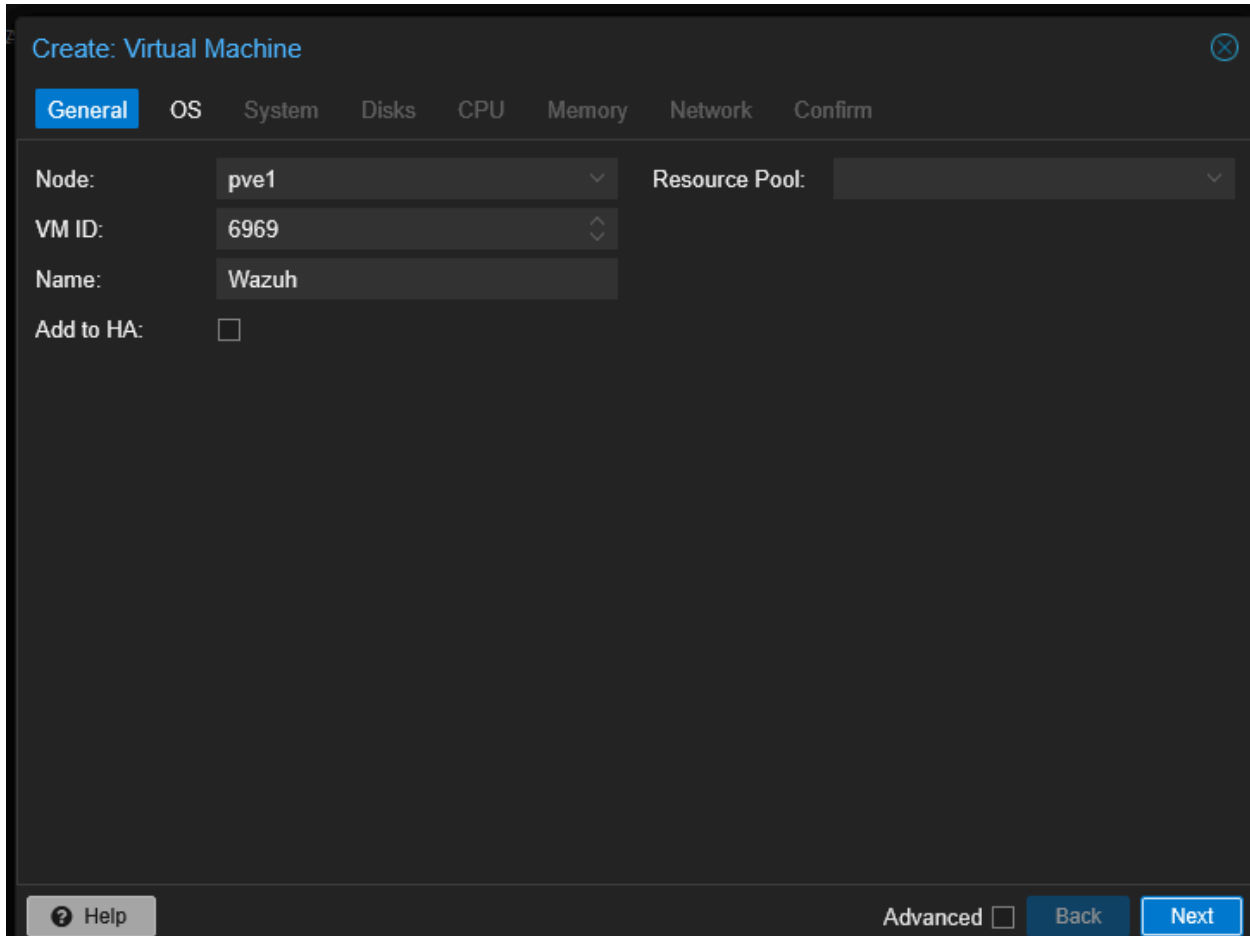


Wazuh Setup Lab

By Keenan Morrison

First I created a VM on Proxmox and gave it a unique VM ID and Name.



The screenshot shows the 'Create: Virtual Machine' window in Proxmox. The 'General' tab is selected, showing fields for 'Node' (pve1), 'VM ID' (6969), 'Name' (Wazuh), and 'Resource Pool'. There is an 'Add to HA' checkbox which is unchecked. At the bottom, there are buttons for 'Help', 'Advanced' (unchecked), 'Back', and 'Next'.

Figure 1.1 Shows me using the VM ID 6969 and the Name Wazuh for my new VM on Proxmox

After that I had to add the OS but the OS I had downloaded wasn't showing up on the OS tab and therefore I have to move the Ubuntu ISO to a the directory that Proxmox uses for storing ISOs so that I could use the ISO for this VM.

```
root@pve1:/# ls
bin boot dev etc home lib lib64 lost+found media mnt opt proc root run sbin srv sys tmp usr var
root@pve1:/# cd root
root@pve1:~# ls
ubuntu-22.04.5-live-server-amd64.iso
root@pve1:~# mv ubuntu-22.04.5-live-server-amd64.iso /var/lib
lib/ local/ lock/ log/
root@pve1:~# mv ubuntu-22.04.5-live-server-amd64.iso /var/lib/vz/template/iso/
root@pve1:~# ls
root@pve1:~# cd
```

Figure 1.2 : Shows the process of moving the ISO to the proper directory.

Then I went to the “Disks” tab and I made sure to configure the disk size to be 50GB.

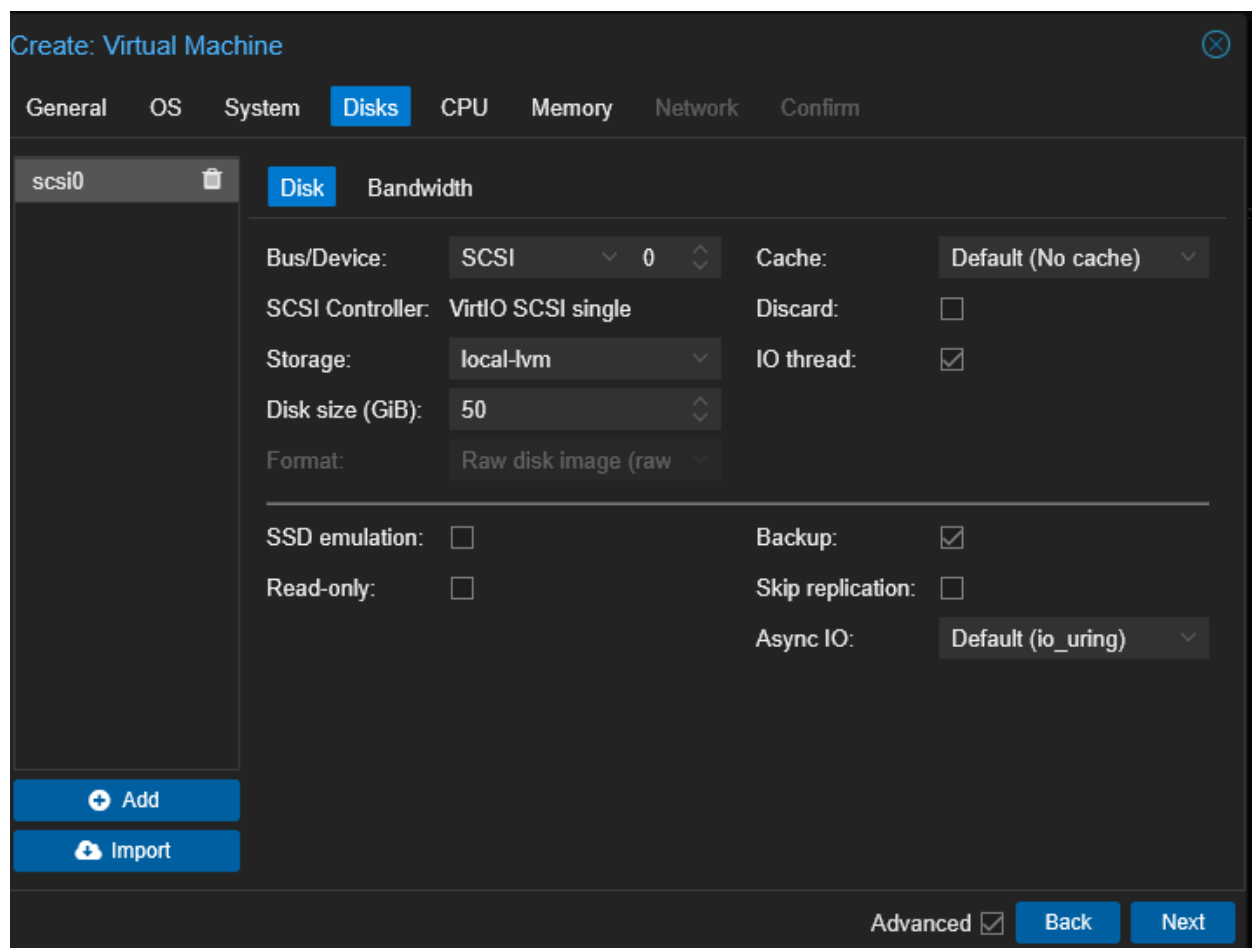


Figure 1.3 : Shows 50 GiB of Storage for Disk Size.

This is what it looked like at the end of configuring the settings for the VM

Create: Virtual Machine ⓧ

General OS System Disks CPU Memory Network **Confirm**

Key ↑	Value
cores	2
cpu	x86-64-v2-AES
ide2	local:iso/ubuntu-22.04.5-live-server-amd64.iso,media=cdrom
memory	5068
name	Wazuh
net0	virtio,bridge=vibr0,firewall=1
nodename	pve1
numa	0
ostype	l26
scsi0	local-lvm:50,iothread=on
scsihw	virtio-scsi-single
sockets	1
vmid	6767

☐ Start after created

Advanced ☒ **Back** **Finish**

Figure 1.4 : Summary of VM configurations

Then I went through the installer for Ubuntu Server until I got to this

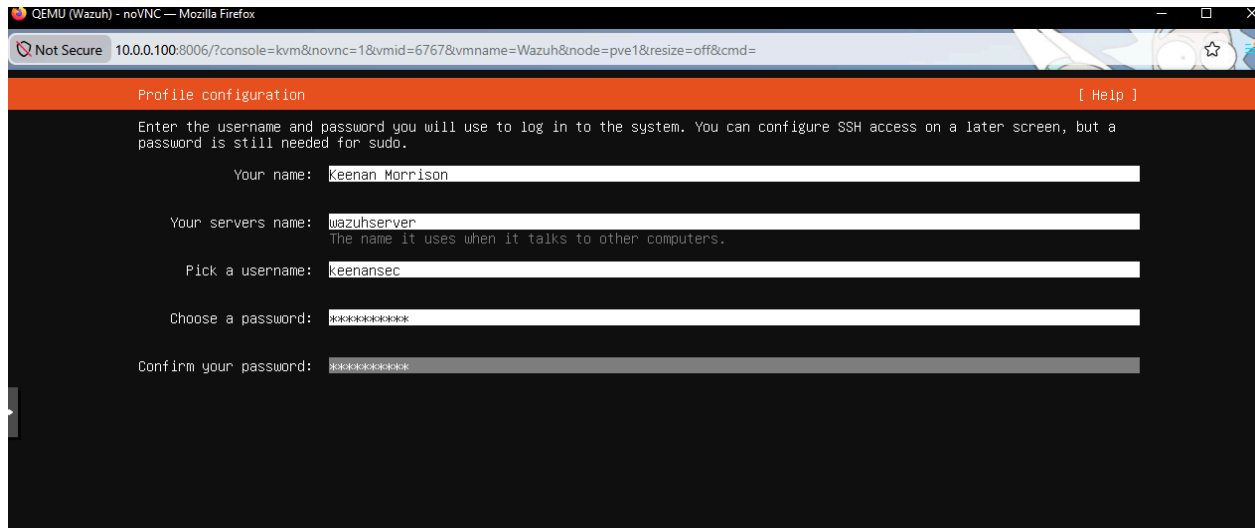


Figure 1.5 : Profile Configuration right before system reboot and initial login

Then I ssh'd into the server and installed the script and config file for Wazuh. After that I changed the IP in "config.yml" to that of the Ubuntu Servers IP. Finally I ran the install script as root and after it gave me the username and password for the login panel.

```
keenansec@wazuhsrver:~$ sudo !!
sudo curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
[sudo] password for keenansec:

keenansec@wazuhsrver:~$ sudo curl -sO https://packages.wazuh.com/4.7/wazuh-inst
all.sh
keenansec@wazuhsrver:~$ curl -sO https://packages.wazuh.com/4.7/config.yml
keenansec@wazuhsrver:~$ nano config.yml
keenansec@wazuhsrver:~$ bash wazuh-install.sh -a
wazuh-install.sh: line 2585: /var/log/wazuh-install.log: Permission denied
This script must be run as root.
keenansec@wazuhsrver:~$ sudo !!
sudo bash wazuh-install.sh -a
11/05/2026 10:56:43 INFO: Starting Wazuh installation assistant. Wazuh version:
4.7.5
11/05/2026 10:56:43 INFO: Verbose logging redirected to /var/log/wazuh-install.l
og
11/05/2026 10:56:51 INFO: Wazuh web interface port will be 443.
11/05/2026 10:56:55 INFO: --- Dependencies ---
11/05/2026 10:56:55 INFO: Installing apt-transport-https.
11/05/2026 10:57:03 INFO: Wazuh repository added.
11/05/2026 10:57:03 INFO: --- Configuration files ---
11/05/2026 10:57:03 INFO: Generating configuration files.
11/05/2026 10:57:04 INFO: Created wazuh-install-files.tar. It contains the Wazuh
cluster key, certificates, and passwords necessary for installation.
11/05/2026 10:57:04 INFO: --- Wazuh indexer ---
11/05/2026 10:57:04 INFO: Starting Wazuh indexer installation.

11/05/2026 10:59:28 INFO: Wazuh indexer installation finished.
11/05/2026 10:59:28 INFO: Wazuh indexer post-install configuration finished.
11/05/2026 10:59:28 INFO: Starting service wazuh-indexer.
11/05/2026 10:59:41 INFO: wazuh-indexer service started.
11/05/2026 10:59:41 INFO: Initializing Wazuh indexer cluster security settings.
11/05/2026 10:59:52 INFO: Wazuh indexer cluster initialized.
11/05/2026 10:59:52 INFO: --- Wazuh server ---
11/05/2026 10:59:52 INFO: Starting the Wazuh manager installation.
11/05/2026 11:01:00 INFO: Wazuh manager installation finished.
11/05/2026 11:01:00 INFO: Starting service wazuh-manager.
11/05/2026 11:01:18 INFO: wazuh-manager service started.
11/05/2026 11:01:18 INFO: Starting Filebeat installation.
11/05/2026 11:01:29 INFO: Filebeat installation finished.
11/05/2026 11:01:30 INFO: Filebeat post-install configuration finished.
11/05/2026 11:01:30 INFO: Starting service filebeat.
11/05/2026 11:01:31 INFO: filebeat service started.
11/05/2026 11:01:31 INFO: --- Wazuh dashboard ---
11/05/2026 11:01:31 INFO: Starting Wazuh dashboard installation.
11/05/2026 11:02:40 INFO: Wazuh dashboard installation finished.
11/05/2026 11:02:40 INFO: Wazuh dashboard post-install configuration finished.
11/05/2026 11:02:40 INFO: Starting service wazuh-dashboard.
11/05/2026 11:02:41 INFO: wazuh-dashboard service started.
11/05/2026 11:03:07 INFO: Initializing Wazuh dashboard web application.
11/05/2026 11:03:07 INFO: Wazuh dashboard web application initialized.
11/05/2026 11:03:07 INFO: --- Summary ---
11/05/2026 11:03:07 INFO: You can access the web interface https://<wazuh-dashbo
ard-ip>:443
User: admin
Password:
11/05/2026 11:03:07 INFO: Installation finished.
keenansec@wazuhsrver:~$
```

Figure 1.6 : Command line history of getting the config file and install script and running the installer script

After this I went to the `http://[IP-ADDRESS]` and used the credentials given to me in the command line

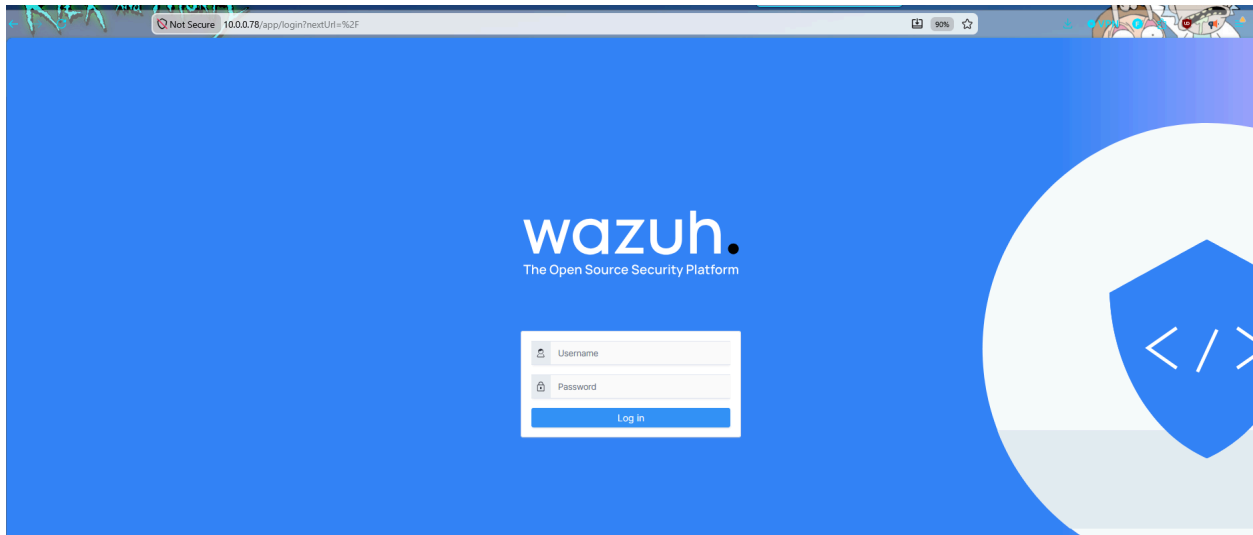


Figure 1.7 : Wazuh Login Screen

I then pressed Deploy new agent and changed the server address to the one of my Ubuntu Server.

Refresh

Deploy new agent

✓

Select the package to download and install on your system:

 **LINUX**

☐ RPM amd64 ☐ RPM aarch64
☐ DEB amd64 ☐ DEB aarch64

 **WINDOWS**

☒ MSI 32/64 bits

 **macOS**

☐ Intel
☐ Apple silicon

For additional systems and architectures, please check our documentation.

✓

Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FDQN).

Assign a server address:

10.0.0.78

✓

Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name:

Juggernaut

The agent name must be unique. It can't be changed once the agent has been enrolled.

Select one or more existing groups:

Default

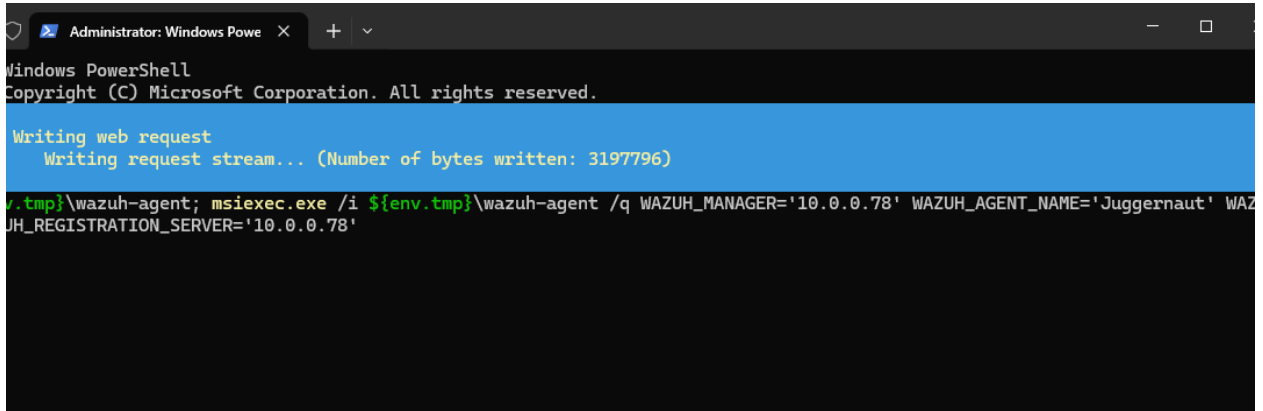
4

Run the following commands to download and install the agent:

```
Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.7.5-1.msi -OutFile  
${env:tmp}\wazuh-agent; msixec.exe /i ${env:tmp}\wazuh-agent /q WAZUH_MANAGER='10.0.0.78'  
WAZUH_AGENT_NAME='Juggernaut' WAZUH_REGISTRATION_SERVER='10.0.0.78'
```

Figure 1.8 : Deploy New Agent Instructions

I then ran powershell as administrator and then copied and run the command show in step 4.



```
Administrator: Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Writing web request
Writing request stream... (Number of bytes written: 3197796)

y.tmp}\wazuh-agent; msexec.exe /i ${env.tmp}\wazuh-agent /q WAZUH_MANAGER='10.0.0.78' WAZUH_AGENT_NAME='Juggernaut' WAZUH_REGISTRATION_SERVER='10.0.0.78'
```

Figure 1.9 : Powershell downloading file via command copied and pasted from Wazuh

Then I ran the command “NET START WazuhSvc” and I seen the device added as an agent on the Wazuh Overview

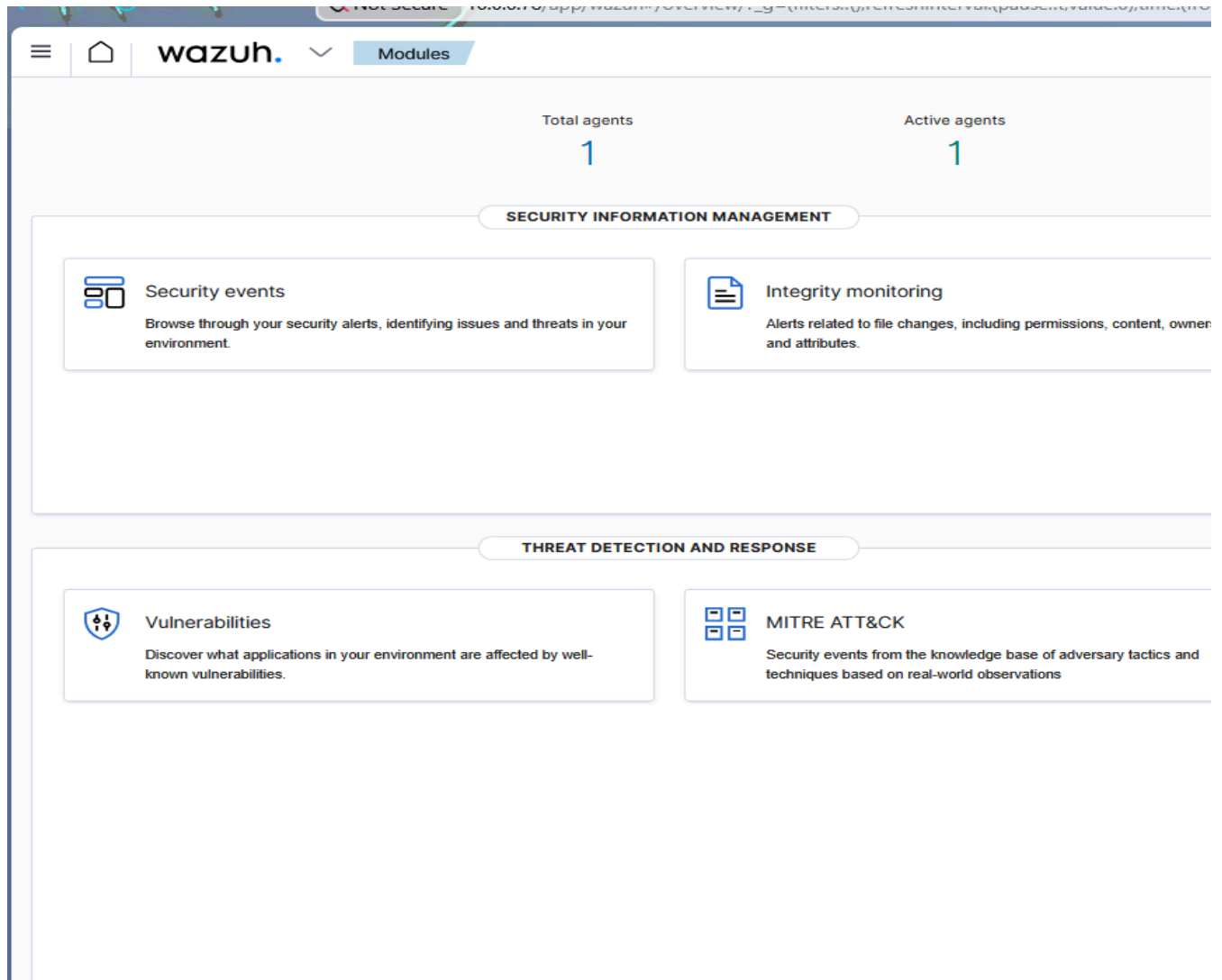


Figure 1.10 : Shows that the Windows machine was successfully added as an agent

What I Learned

- **Ubuntu Server Configuration**
- **Wazuh Server Setup**
- **Wazuh Agent Setup**
- **How to create VMs on Proxmox**
- **How to move ISOs that are downloaded and not listed on Proxmox to be listed**